

Problem 10

If any one of four products is defective, then this is event A ; if no fewer than two products are defective, then this is event B . What do the complementary events $\bar{A}$ and $\bar{B}$ mean?

Solution

We need to interpret events A and B , then determine their complements.

Step 1: Clarify Event A

Original Statement

"If any one of four products is defective, then this is event A "

The phrase "any one" is ambiguous. It could mean:

- **Interpretation 1:** "At least one product is defective" (any = at least one)
- **Interpretation 2:** "Exactly one product is defective"

Given the context and the comparison with event B ("no fewer than two"), Interpretation 1 is more natural.

Event A Definition

$A =$ "At least one of the four products is defective"

This means: 1, 2, 3, or 4 products are defective.

Step 2: Clarify Event B

Statement

"If no fewer than two products are defective, then this is event B "

"No fewer than two" means "at least two" or "two or more."

Event B Definition

$B =$ "At least two of the four products are defective"

This means: 2, 3, or 4 products are defective.

Step 3: Relationship Between A and B

Let k denote the number of defective products among the four.

Number of defects	Event A	Event B
$k = 0$	No	No
$k = 1$	Yes	No
$k = 2$	Yes	Yes
$k = 3$	Yes	Yes
$k = 4$	Yes	Yes

From this table, we see that $B \subset A$ (every outcome in B is also in A).

Step 4: Determine $\bar{A}$ (Complement of A)

Definition of Complement

The complement $\bar{A}$ consists of all outcomes not in A .

What is NOT in A ?

Event A = "at least one product is defective" = " $k \geq 1$ "

The complement is:

$\bar{A}$ = "NOT (at least one product is defective)"

= "No products are defective"

= "All four products are functional"

Answer for $\bar{A}$

$\bar{A}$ = "None of the four products is defective (all are functional)"

Equivalently: $k = 0$ (zero defective products).

Step 5: Determine $\bar{B}$ (Complement of B)

What is NOT in B ?

Event B = "at least two products are defective" = " $k \geq 2$ "

The complement is:

$\bar{B}$ = "NOT (at least two products are defective)"

= "Fewer than two products are defective"

= "At most one product is defective"

= "Zero or one product is defective"

Answer for $\bar{B}$

$\bar{B}$ = "At most one product is defective (0 or 1 defective)"

Equivalently: $k \in \{0, 1\}$.

Step 6: Summary Table

Number of defects	A	B	$\bar{A}$	$\bar{B}$
$k = 0$	No	No	Yes	Yes
$k = 1$	Yes	No	No	Yes
$k = 2$	Yes	Yes	No	No
$k = 3$	Yes	Yes	No	No
$k = 4$	Yes	Yes	No	No

Step 7: Verification

Check: $A \cup \bar{A} = \Omega$ (entire sample space)

- A : $k \in \{1, 2, 3, 4\}$
- $\bar{A}$: $k \in \{0\}$
- $A \cup \bar{A}$: $k \in \{0, 1, 2, 3, 4\}$

Check: $A \cap \bar{A} = \emptyset$

No value of k satisfies both A and $\bar{A}$ simultaneously.

Check: $B \cup \bar{B} = \Omega$

- B : $k \in \{2, 3, 4\}$
- $\bar{B}$: $k \in \{0, 1\}$
- $B \cup \bar{B}$: $k \in \{0, 1, 2, 3, 4\}$

Check: $B \cap \bar{B} = \emptyset$

No value of k satisfies both B and $\bar{B}$ simultaneously.

Step 8: Set Relations

From the analysis:

- $B \subset A$ (if at least 2 are defective, then at least 1 is defective)
- $\bar{A} \subset \bar{B}$ (if 0 are defective, then at most 1 is defective)
- $\bar{B} = \bar{A} \cup (A \cap \bar{B})$ where $A \cap \bar{B}$ is the event "exactly one is defective"

Step 9: In Terms of Set Operations

We can write:

$$A = \{k = 1\} \cup \{k = 2\} \cup \{k = 3\} \cup \{k = 4\}$$
$$\bar{A} = \{k = 0\}$$

$$B = \{k = 2\} \cup \{k = 3\} \cup \{k = 4\}$$
$$\bar{B} = \{k = 0\} \cup \{k = 1\}$$

Final Answer

Complement of Event A :

$\bar{A} = \text{"None of the four products is defective"}$

Meaning: All four products are functional (zero defects).

Complement of Event B :

$\bar{B} = \text{"At most one product is defective"}$

Meaning: Either all four products are functional, or exactly one product is defective (0 or 1 defects).

In Summary:

- A : At least 1 defective $\implies \bar{A}$: Exactly 0 defective
- B : At least 2 defective $\implies \bar{B}$: At most 1 defective (0 or 1)

Set Relationship: $\bar{A} \subset \bar{B} \subset A^c \cup B^c = \bar{A} \cup \bar{B}$

Actually, more precisely: $\bar{A} \subset \bar{B}$ because if no products are defective, then certainly at most one is defective.